

Identifying Junk Electronic Mail in Microsoft Outlook with a Support Vector Machine

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Abstract

In this paper, we utilize a simple support vector machine to identify commercial electronic mail. The use of a personalized dictionary for model training provided a classification accuracy of 96.69%, while a much larger system dictionary achieved 95.26%. The classification system was subsequently implemented as an add-in for Microsoft Outlook XP, providing sorting and grouping capabilities using Outlook's interface to the typical desktop e-mail user.

1. Introduction

Unsolicited commercial electronic mail (UCE) has plagued Internet users for almost an entire decade. The first recognized incident of large-scale mass mailing occurred in 1994, when two California attorneys sent their green card solicitation to most of USENET [1]. Since that time, bulk mailing software has proliferated, making it possible to transmit thousands of messages with forged headers and falsified contact information without much technical knowledge.

Unwanted e-mail is generally categorized under the blanket term spam, although a casual analysis suggests that multiple types exist. It is particularly important to distinguish between solicited commercial e-mail and UCE. Solicited commercial e-mail is typically generated by a large corporation in response to a request for additional periodic information. Because they are authored by corporations which tend to be cognizant of their customers' privacy and complaints, solicited commercial e-mail can typically be abated through an unsubscribe process. Occasionally, corporate e-mail may be solicited for delivery to a third party recipient. These messages, such as online greeting card notifications and "tell a friend about this page" referral services, generate a nebulous form of UCE which contain no personal communications. Most spam, however, generally lacks any corporate legitimacy, and typically advertises a wide range of products and services including office and computer products, adult entertainment products, make-money-fast (MMF) opportunities, and pyramid schemes.

In the past, spam filtering required the manual construction of pattern matching rule sets [1]. This process was not particularly efficient, catching only a small portion of incoming UCE and requiring extensive amounts of time. Probabilistic classification techniques simplify the filtering process by providing automated classification model generation based on representative samples. In [5], Sahami presented one of the first applications of probabilistic text classification to label electronic mail as spam or nonspam. By using simplified factor analysis to select 500 words as features for classification, Sahami was able to produce accuracy results between 87.7% and 97.1% with a Bayes classifier. In [2], Drucker applied a linear support vector machine (SVM) with the sequential minimal optimization (SMO) training method [4] to the task of classifying electronic mail, comparing the performance with other techniques. Drucker concluded that SVMs produced some of the best results when working with binary features representing all words in the feature space.

The purpose of this project is to use a SVM to construct an automated classification system to detect unsolicited commercial e-mail. Several sets of sample messages were collected to build dictionaries of words found in e-mail communications, which were processed by the SVM to create a classification model for spam or nonspam messages. This model was utilized in an add-in for Microsoft Outlook XP, allowing desktop mail users to quickly identify unsolicited electronic mail.

2. Background

In text categorization with a SVM, a word is a feature. The list of all the words recognized by the system, that is, the dictionary, is called the feature space. An e-mail message may be represented by a feature vector $\mathbf{x}$, containing elements x_i which correspond to the each word in the dictionary. In the binary representation, a feature vector element is set to one if the word was present in the e-mail and zero if it was not. A SVM trainer, such as Joachim's svmLight [3] or Platt's SMO [4], examines a series of training vectors to determine the location of a hyperplane that best separates the samples of the different categories. The training process produces a weight vector

$\mathbf{w}$ with elements corresponding to those in the feature vector $\mathbf{x}$. In this application, positive weights indicate that the particular feature is associated with a spam message, while negative weights indicate association with a nonspam message.

Any e-mail message may be classified as spam or nonspam by performing a simple dot product between the message's feature vector and the SVM model weight vector,

$$u = \mathbf{w} \cdot \mathbf{x} - b,$$

where u is the result of the classification and b is the SVM model's bias parameter determined by the training process. Positive values for u indicate spam messages and negative values indicate nonspam messages, with larger values representing stronger characteristics of the classification. As the feature vector contains binary features, no multiplication is necessary to perform the dot product operation. Instead, the message's classification may be generated by iterating through the feature vector and simply summing the weights of any words that were found in the message.

3. Data Sets

For the purposes of training and testing the classification SVM, several sets of sample electronic mail were collected. Over a two month period, individuals at RIT's Information Technology Services help desk archived 1342 messages which were evaluated by the recipient as junk mail. These messages were combined with 1342 nonspam personal messages collected by the author over a period of two years. The combined e-mail data set, which contained 2684 messages, was randomly divided into a training set and a testing set containing 1340 and 1344 messages, respectively, with an equal number of spam and nonspam messages. The training set was used only for dictionary generation and SVM model creation, while the test set is referred to as Set A.

To assist in evaluating the generalization performance of the classifier, an additional data set was collected and used for testing. This second set, referred to as Set B, contains 77 spam messages collected by the author and 378 nonspam messages collected by a third party. The intent is to demonstrate that the filter constructed with other people's data can be applied impersonally; that is, without building a specific model for a particular recipient.

4. Method

Each mail message was parsed to completely remove any headers, attachments, HTML markup, punctuation, and extended characters. This procedure essentially reduces a mail message to a series of delimited lowercase

string tokens. The messages in the training set were used to generate a list containing 19,108 tokens. This raw list was processed by removing any token which occurred less than three times. In addition, any tokens beginning with http or mailto, and tokens consisting entirely of numeric digits, were removed. This resulted in a dictionary of 5,841 words. Because the dictionary was generated by an analysis of the personal e-mail of an individual, a significant number of proper nouns were present. This dictionary is referred to as the personal dictionary. As an additional test, the 16 most frequent proper nouns were removed, resulting in a more impersonal dictionary. The removal of proper nouns prevents classification based on names used as closing lines or signatures, forcing the classifier to examine only generic words. Finally, the system wordlist retrieved from a UNIX workstation was utilized as an alternate dictionary.

Each message in the training set was parsed to generate feature vectors using the three dictionaries. Unfortunately, a small percentage of e-mail messages did not contain any words in the dictionary. These messages cannot be parsed to generate feature vectors, and are omitted from the subsequent testing results. This rate of omitted messages was typically very small, at 1% on Set A and 7% on Set B, dependent on the contents of the messages. The feature vectors were used to produce a classification model using a SVM trainer product based on Platt's SMO algorithm. With the SVM parameter C set to 10 and a linear kernel, the training time for each model was quite short, requiring less than 2 minutes to complete on a desktop workstation. All three models produced valid solutions with perfect recall accuracies, which were then used to test the remaining data sets.

5. Results

The classification models were quite accurate when performed with the extracted feature set dictionaries. Results for the primary message collection are presented in Table 1. The personal and impersonal dictionaries produced identical results, achieving the system's best accuracy at 96.69%. The system wordlist was slightly less accurate at 95.56%. Results for the third party message collection are presented in Table 2. Overall, the results were slightly less accurate, with the personal and impersonal dictionaries producing similar results and the system dictionary the worst results.

Analysis of the linear SVM models indicates which dictionary words are associated with spam and nonspam messages as well as which words are not significant in determining the classification (see Table 3, Table 4, and Table 5). In Tables 3 and 4, accuracy is reported as the overall percentage of messages classified correctly. The false alarm rate describes the percentage of nonspam

messages incorrectly classified as spam, while the miss rate indicates the percentage of spam messages incorrectly classified as nonspam.

The personal and impersonal dictionaries, generated through a dictionary-building process based on sample messages, resulted in words which were roughly balanced between nonspam, neutral, and spam categories. The system dictionary, however, had the fewest number of words in its classification categories, which may explain its lesser performance.

Table 1 – Classification results for primary message collection (Set A)

Dictionary	Accuracy	False Alarms	Misses
Personal	96.69%	4.52%	2.10%
Impersonal	96.69%	4.52%	2.10%
System	95.26%	6.80%	2.70%

Table 2 – Classification results for third party message collection (Set B)

Dictionary	Accuracy	False Alarms	Misses
Personal	94.23%	6.42%	2.67%
Impersonal	94.23%	6.15%	4.00%
System	90.30%	9.50%	10.67%

The personal dictionary produced a model with large nonspam weights for proper nouns associated with the author. These nouns included both first and last name, academic institution, and the names of former roommates. When these names were removed from the dictionary, and were therefore not available for classification, accuracy did not change substantially. These results suggest that a personalized dictionary including proper nouns may not be particularly important and that a well-developed generic dictionary may be applied at the server level to an incoming mail stream.

Table 3 – Trained dictionary statistics

Dictionary	Words	Nonspam	Neutral	Spam
Personal	5841	1888	2172	1781
Impersonal	5825	1838	2131	1856
System	25473	1192	23331	950

While the selection of words is important, the presence of words is essential. In order for the SVM to be able to generate a classification, the message must first be parsed into tokens and correlated with the dictionary to produce a feature vector. This seems intuitive, but electronic mail messages do not need to contain text. In the sample sets used for the training and testing of this classifier, several messages were discarded because they contained no words which were present in the dictionary. These messages may fall into three categories. First, the message may be blank, with the entire content of the transmission included in the subject header. Second, the message may actually not contain any words. These

messages are typically generated by the sender intentionally pasting a URL into an e-mail message with no additional conversation. Third, the message may not contain any words due to image-only HTML markup. While the previous two unclassifiable message types may be spam or nonspam, this final category is a feature generally only present in junk mail. Individuals typically type messages directly into an e-mail client, but some spam authors present their advertisement only as JPG or GIF graphics referenced using HTML image tags. The feature extraction parsing sequence eliminates all HTML tags, which leaves the document blank. This type of message is particularly problematic, as the transmission contains no ASCII text except for the headers which may be used to produce an automated classification.

6. Microsoft Outlook Integration

To provide the benefits of SVM-based e-mail classification to ordinary end users, an add-in DLL for Microsoft Outlook XP was developed. This software examines electronic mail messages on delivery, running each message delivered to the default Inbox through feature extraction and SVM classification stages. Written entirely in Microsoft Visual Basic, the spam scanner add-in implements the extractor and SVM internally, and interfaces with Outlook XP using its published object model.

When new mail is received, the spam scanner automatically loads and begins the classification process. The results of the spam classification are stored with the Microsoft Outlook mailbox item using a custom property named SPAM. This custom property becomes a field associated with the information store, so it is available to the Outlook environment using the traditional user interface. Thus, the SPAM field may be displayed as a column in the view of e-mail messages and used as criteria for conditional formatting. For example, messages that are labeled spam may be displayed in a different color than the legitimate messages, or used to group potentially unsolicited messages at the bottom of the Inbox.

7. Conclusions

Although the classification system obtained a high accuracy, certain types of messages are difficult to classify using textual features alone. These messages include automatically generated newsletters, online store advertisements, and greeting card notifications. To avoid the loss of potentially useful messages, it is important to use the message content based SVM classification system only as a labeling and prioritizing assistant, instead of deleting messages solely on its judgment.

Misclassifications are generally due to solicited commercial messages which contain many of the features of spam, including automatic generation, lack of personalized content, and verbose unsubscribe notices. Ultimately, one person's spam may be another person's subscription, so a certain genre of messages will require manual intervention either interactively or through the construction of traditional domain-specific processing rules.

This initial work suggests several areas for future research. First, and most significant, is additional formal testing to determine any statistical correlation between classifier accuracy and the use of a personalized dictionary. Unfortunately, this is difficult because most individuals are hesitant to release archives of personal or business e-mail messages for analysis despite guarantees of privacy and anonymity through preprocessing. Second, this SVM model's classification accuracy should be compared with results from several current commercial spam filtering products. Finally, the classification system may be more useful when implemented as server-side software with appropriate user interfaces for domain-specific exceptions and incremental retraining.

The electronic mail classification application provides a highly efficient and accurate method of prescreening unsolicited e-mail. Without generating any custom rules, such as exceptions for preferred subscriptions, the system's accuracy exceeded 96%. In addition to the experimental setup utilized for generating and testing the models, the Microsoft Outlook add-in allows ordinary users to utilize SVM technology to find and discard mailbox junk.

8. References

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Table 4 – Words indicating nonspam messages

Word	Weight	Word	Weight
i	-0.3565	course	-0.1236
do	-0.3038	its	-0.1230
e	-0.2642	hello	-0.1222
read	-0.2453	how	-0.1217
go	-0.2428	surprise	-0.1215
would	-0.2045	what	-0.1213
that	-0.1858	he	-0.1195
mom	-0.1785	still	-0.1184
pm	-0.1749	think	-0.1183
hey	-0.1698	thanks	-0.1179
day	-0.1681	am	-0.1155
the	-0.1605	night	-0.1153
page	-0.1520	needs	-0.1141
msn	-0.1479	always	-0.1138
may	-0.1471	hope	-0.1134
when	-0.1347	almost	-0.1131
hi	-0.1308	will	-0.1129
then	-0.1303	documentation	-0.1120
im	-0.1290	know	-0.1108
thursday	-0.1283	note	-0.1087
dad	-0.1275	virtual	-0.1084
are	-0.1251	questions	-0.1080
but	-0.1246	some	-0.1070

Table 5 – Words indicating spam messages

Word	Weight	Word	Weight
mailing	0.1522	these	0.2052
action	0.1523	future	0.2096
watch	0.1528	hot	0.2164
from	0.1542	shipping	0.2172
remove	0.1563	adult	0.2172
offers	0.1568	at	0.2189
internet	0.1578	sex	0.2309
money	0.1613	must	0.2329
want	0.1647	be	0.2332
access	0.1681	porn	0.2474
incredible	0.1714	more	0.2528
list	0.1728	sites	0.2556
to	0.1792	celebs	0.2706
their	0.1823	best	0.2718
very	0.1837	removed	0.2731
take	0.1861	tight	0.3003
get	0.1874	unsubscribe	0.3032
below	0.1874	click	0.3148
no	0.1888	teens	0.3563
handle	0.1888	hardcore	0.3907
show	0.1889	young	0.4131
now	0.1910	here	0.4850
mailings	0.2008	free	0.5452